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ORIGINAL RESEARCH

Kinematic control in a four-wheeled Mecanum mobile robot for trajectory tracking

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Abstract

The challenges of the modern world require mobile robots with the ability to navigate in congested environments with high levels of manoeuvrability. Therefore, the Mecanum wheel may be viable for addressing this challenge. This paper presents the experimental results of a kinematic control strategy, which involves considering the dynamic model as a black box, with only the input and output signals being known. To do this, high-level and low-level controllers are formulated and explained. The high level aims to control the desired position of the mobile robot, which can be useful for navigation tasks. On the other hand, low-level control involves nested controllers to regulate the speed of the mobile robot wheels. Both levels are related and computed through pure kinematics transformations with a dSPACE ds1103 card and MATLAB/Simulink software. In total, 120 experiments were conducted to determine the repeatability of the tests, using the combination of three widely explored control techniques in the literature: proportional-integral-derivative (PID), PID plus sliding modes, and PID plus quasi-sliding modes. The experiments conducted are described in detail, and the results are analysed using statistical indices based on the RMS error and percentage improvement.

1 | INTRODUCTION

The challenges of the modern world in wheeled mobile robotics require solutions that ensure high manoeuvrability and flexibility, to cover a wide range of applications [1, 2]. To address these challenges, the Mecanum wheel, invented by Bengt Illon in 1973 [3], is a relevant option in wheeled mobile robots. The Mecanum wheel is composed of passive rollers made of a rubber alloy, fixed onto the wheel, mounted at a 45° angle concerning the wheel's axis of rotation. There are two types of Mecanum wheels: those with rollers oriented at +45° and those oriented at −45° [4–6]. Due to the rollers, the mobile robot can move in any direction: forward, backward, right, left, diagonally at any angle, and rotate on its axis, all while maintaining its initial pose and by that, the Mecanum wheel belongs under the classification of omnidirectional wheels [7]. This type of robot is increasingly being explored by companies, scientists, engineers, students, and enthusiasts for their unique movement capabilities. Some of the main applications of this class of robots are smart manufacturing [8], heavy-duty material handling [9, 10], healthcare, where they have been used for medication delivery and distribution [11], Mecanum wheels have also been used in electric wheelchairs for people with disabilities to allow them to have greater flexibility in their movements [12, 13]. Other applications can be found in the systematic survey presented in [14].

Many control approaches have been formulated for Mecanum-wheeled mobile robots to allow them to solve position control. Most control law designs are model-based control; this approach involves formulating its dynamic model, which is obtained using Lagrangian formalism [15, 16]. Most of the strategies in the literature delve into robust control to address undesirable effects in practice, such as external disturbances, parametric uncertainties, and surface irregularities. Subsequently, those are compensated by a control law. Sliding-mode control-based techniques [17–19], and backstepping control [20, 21] are techniques found in the literature where robust control is applied to mobile robots with Mecanum wheels using a model-based approach. For example, in [22], an extended state-observer based on sliding mode control (ESO-SMC) is presented to estimate the unmeasurable state and the lumped disturbances/uncertainties term based on a dynamic-model controller. The comparative results illustrate that the

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ESO-SMC controller has better tracking performance than the standard PID controller. Another example is presented in [23], where a sliding mode control law is proposed for trajectory tracking in a Mecanum-wheeled mobile robot. Due to the effects of chattering, the sign function is replaced by a saturation function. In addition, an adaptive control law is implemented to handle abrupt turns. The results showed good performance with minimal error and cancellation of chattering. Concerning intelligent control methods, neural network algorithms and fuzzy logic control are presented in [24] and [25], respectively. Also, it is important to note that it has been observed that mobile robots equipped with Mecanum wheels have a greater tendency to random slippage, as opposed to other wheel types. The slippage causes unpredictable deviations in the orientation of the mobile robot, which is undesirable. By that, in [26], an adaptive heading correction controller is presented based on a PID controller with a Kalman filter responsible for updating the vehicle's pose. The results achieved in one of its experiments showed an improvement from 4.56° to 0.38°. Although it does not address a position control problem and only performs wheel velocity control for basic movements such as forward or sideways, discussing slippage in robots using Mecanum wheels is of particular interest and must be considered, as it can lead to position errors [27].

However, it is worth saying that from an experimental and a more practical perspective, the challenge is to design controllers that can be easily implemented and react appropriately to different external disturbances while maintaining good tracking accuracy and robustness. Thus, an important aspect from model-based control is the need for precise parameters of the robot's dynamic model, such as mass, friction coefficients, and inertia, which are difficult to obtain in practice. Also, control laws derived from model-based methods involve solving complex differential equations, which can increase computation time and computational requirements. Therefore, it is important to note that with the kinematic model of a wheeled mobile robot, we establish the relation of the robot's global velocities with the wheel's angular velocities. Under that criterion, measuring the wheel's angular velocity, computing kinematic transformation, and integration process could be sufficient for designing control algorithms based on the computation of direct and inverse kinematics in an embedded system. This approach is called kinematic or motion control [28]. Motion control is used for several kinds of mobile robots. It involves considering the dynamical model of the mobile robot as a black box, in which only the inputs and output signals are known. The kinematic control approach applies similarly to any robot, regardless of its configuration or wheel type. For example, in the work of [29], a kinematic controller is presented for a differential-drive mobile robot and presents a continuous approximation of the sliding mode, which the author referred to as pseudo-sliding mode. This version was later discretized and implemented. Their results showed good performance in trajectory tracking compared to a standard PID control, achieving better settling times and a percentage improvement of 16.35% and 37.26% on the X and Y axes, respectively. The most basic process for achieving navigation in a mobile robot is based on its kinematic model. Therefore, in [30], a kinematic control is presented for a differential drive robot, where the robot's estimation is achieved through odometry. Their experimental results demonstrate repeatability in its implementation and a minimal deviation (in degrees), with an average of 5.5° for distances travelled of 80 cm. Regarding mobile robots with differential locomotion, In [31], motion control is presented based on pure kinematics, where odometry techniques and PID control for robot position and motor speed were employed. The experiment setup was made for square- and triangle-shaped trajectory tracking. The results show robot steady-state errors in the desired trajectories of 10.4 cm for the X-axis and 12.24 cm for the Y-axis for the square shape, while errors of 22.1 cm and 12.3 cm are obtained for the triangle shape on the X-axis and Y-axis, respectively. Thus, kinematic control is practical and helpful; however, it is worth noting that a pure PID-based control method cannot consider the effects of uncertainties due to the surface. Then, the challenge is to implement a robust control law with a black-box dynamic model and react appropriately to uncertainties.

In this paper, we propose a strategy for controlling a four-wheeled Mecanum mobile robot based on information obtained purely from its kinematic model, where the robot's dynamic model is considered a black box, with only the input and output signals being well-known. Based on this, two levels of control are formulated; they are obtained from kinematic transformations called high-level and low-level control. First, high-level control refers to controlling the robot's position regarding its Cartesian coordinates and pose or orientation angle. On the other hand, kinematics transformations can determine the angular velocity the wheels require to reach a specific global position. In this sense, a control method referred to as low-level control is also formulated. It is associated with controlling the angular velocity of the wheels, which are commonly directly coupled to the shaft of a direct current motor; then, the angular velocity can be measured if the motor incorporates an encoder. The terms 'high-level' and 'low-level' have also been adopted in the works of [32] and [33]. For example, this is akin to discussing Cartesian or joint control of a manipulator robot [34]. The implemented high-level controller is a proportional-integral-derivative (PID) control plus a term given by the continuous approximation of the sign function, known in the literature as quasi-sliding mode, to add robustness to the system in a safe way. The results are compared with a PID control and a PID plus sliding mode controller in its native form. For all cases, for the low-level control, we use a PI for speed control for each wheel.

On the other hand, in this study, we establish the desired orientation angle as zero. Therefore, the robot's pose is assumed to be the same at the start and end of the trajectory. The results of this work are illustrated entirely experimentally in a laboratory environment, where the real-time data acquisition card dSPACE ds1103 is used for control algorithm execution. A total of 120 experiments were conducted using two surfaces with different characteristics. The results regarding the RMS error and the percentage improvement of the techniques are analysed. On the other hand, we establish the desired orientation angle as zero. Therefore, the robot's pose is assumed to be the same at

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the start and end of the trajectory. Then, to illustrate the slippage problem, the performance of the orientation angle of the robot during its displacements is presented, and qualitative and statistical results are explained.

The paper is organized as follows: Section 2 presents a detailed description of the mobile robot platform developed for conducting this study. Section 3 presents the kinematic model of a mobile robot with 4 Mecanum wheels. Section 4 presents the kinematic control, based on high-level and low-level approaches. Section 5 presents the experiment setup, detailing the trajectories references, and velocity and position estimation methods. Section 6 presents the experimental results obtained.

2 | MECANUM-WHEELED MOBILE ROBOT PLATFORM

2.1 | Description of the mobile robot prototype

The prototype consists of a rectangular platform mounted on four identical 12V (DC) motors (Pololu brand model 9.7:1) equipped with a gearbox of straight metal gears and an incremental optical encoder. Through its reducer, each motor is coupled with a Mecanum wheel responsible for providing omnidirectional locomotion to the platform. The L298N driver was used as the motor controller for the speed and direction control of each motor through pulse-width modulation (PWM). The mobile robot incorporates a 3-Space embedded inertial measurement unit (IMU) from YostLabs, which is utilized for vehicle orientation measurement. To power the robot, a rechargeable four-cell 6800 mAh Li-Po battery was installed, with connectors located on the outer contour of the vehicle for recharging. The robot was manufactured using 3D printing and CNC machining. The characteristics of the robot are summarized in Tables 1–4 as follows: Table 1 presents the specifications of the robot platform, Table 2 corresponds to the features of the Mecanum wheel used, Table 3 summarizes the key specifications of the DC motor employed and Table 4 synthesizes the electrical specifications of the robot. Finally, an actual photograph of the mobile robot can be observed in Figure 1.

TABLE 1 Mobile robot specifications.

Parameter	Value
Length [mm]	578
Width [mm]	260
Height [mm]	90
Vehicle weight [kg]	4.7

TABLE 2 Mecanum wheel specifications.

Parameter	Value
Number of wheels	4
Number of rollers per wheel	9
Roller angle [degrees]	45
Wheel diameter [mm]	60
Wheel thickness [mm]	32

TABLE 3 Motor specifications.

Parameter	Value
Number of motors	4
Stall current [A]	1.8
No-load current [mA]	100
Gear ratio	9.7:1
No-load speed [RPM]	800
Extrapolated stall torque [kg cm]	1.7
Maximum power [Watts]	3.4
Nominal voltage operation [V]	12

TABLE 4 Electrical specifications.

	Value
Power supply	
Battery type	Li-Po
Nominal voltage [V]	14.8
Current capacity [mA]	100
Download rate [C]	60
Motor driver	
L298N voltage supply [V]	10 to 30
Number of channels	2
IC logic voltage supply [V]	3.5 to 5
Peak output current per channel [A]	12
Sensors	
Encoder voltage supply [V]	3.5 to 5
Encoder current consumption [mA]	10
IMU 3-Space Embedded voltage supply [V]	3.3 to 6
IMU current consumption [mA]	45

2.2 | Data acquisition scheme and control platforms

The scheme used for data acquisition and control experiments on the mobile robot is explained as follows. First, as the central control platform is the dSPACE ds1103 real-time data acquisition platform, it was used to obtain each Mecanum wheel angular velocity from the encoder reading. Both high-level and low-level control algorithms were also executed using this control platform. The control law signal was also sent from the dSPACE ds1103 to the motor drivers of the mobile robot. The control algorithms were programmed in MATLAB and Simulink, and all the data obtained was managed using ControlDesk software. Secondly, the myRIO 1900 board from National Instruments was used as a secondary platform to get

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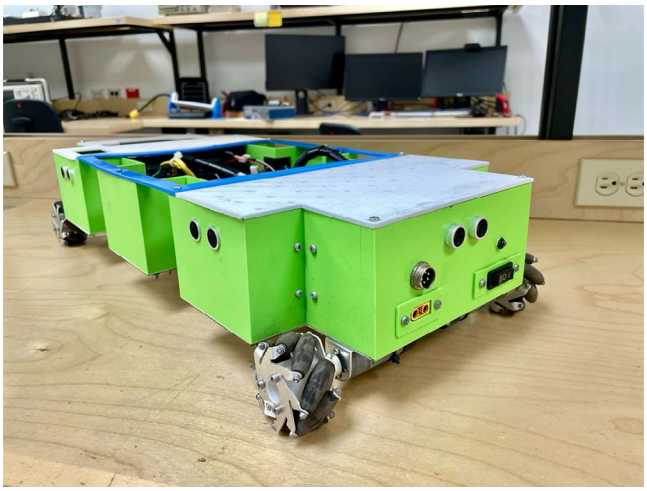


FIGURE 1 Mecanum-wheeled mobile robot used for experimentation.

the vehicle's orientation angle through LabVIEW programming. The orientation angle captured in the myRIO platform is sent to the central control platform via an A/D converter. Figure 2 presents a block diagram illustrating the connections between all these modules.

3 | KINEMATIC MODEL

3.1 | Notation

Figure 3 illustrates the main kinematics entities of the four-wheeled Mecanum mobile robot and is described as follows. Firstly, X_G, G, Y_G is the fixed reference frame of the robot, which also can be interpreted as the initial condition for each experiment. X_R, R, Y_R is the reference frame associated with the mobile robot's center of mass. V_{XR} and V_{YR} [m/s] are the global linear velocities of the robot. $\theta_{\hat{z}}$ [rad/s] is the global angular velocity of the robot. X_r, r, Y_r is a reference frame for each Mecanum wheel. l_x and l_y [m] are the distance between the wheels' geometric centers along the platform's width and length. V_{ir} for $i = 1, \ldots, 4$. is the velocity vector generated by the roller's contact with the surface. γ is the angle of the V_{ir} vector respect with $X_{r,i}$ axis. θ [rad] is the measured angle, which denotes the robot's pose and is expressed as the angle of X_R axis with respect to the fixed reference axe X_G. Before kinematic formulation, the following assumptions are established.

Assumption 1. To formulate the kinematic model of a mobile robot with four Mecanum wheels, the surface is assumed to be free of irregularities and inclinations, and slippage is not a present phenomenon.

Assumption 2. The robot structure is an ideally rigid body whose parameters l_x, l_y, and r do not change over time.

3.2 | Formulation

The model presented below has been extensively explored in the works of [35] and [36]. However, a detailed explanation is provided. We begin by describing the velocity of the ith wheel in terms of its associated reference frame X_r, r, Y_r:

$$V_{Xi} = V_{r,i} - V_{ir}\cos(\gamma_i) = r\omega_i - V_{ir}\cos(\gamma_i), \tag{1}$$

$$V_{Yi} = V_{ir}\sin(\gamma_i), \tag{2}$$

where $r\omega_i$ is the angular velocity of the wheel, and the terms $V_{ir}\cos(\gamma_i)$, $V_{ir}\sin(\gamma_i)$ are the roller velocity decomposition components along the x-axis and y-axis, respectively, of the wheel reference frame X_r, r, Y_r. The above equations can be represented in a compact matrix form:

$$\begin{bmatrix} V_{Xi} \\ V_{Yi} \end{bmatrix} = \begin{bmatrix} r & -\cos(\gamma_i) \\ 0 & \sin(\gamma_i) \end{bmatrix} \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix} = T_w \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix}, \tag{3}$$

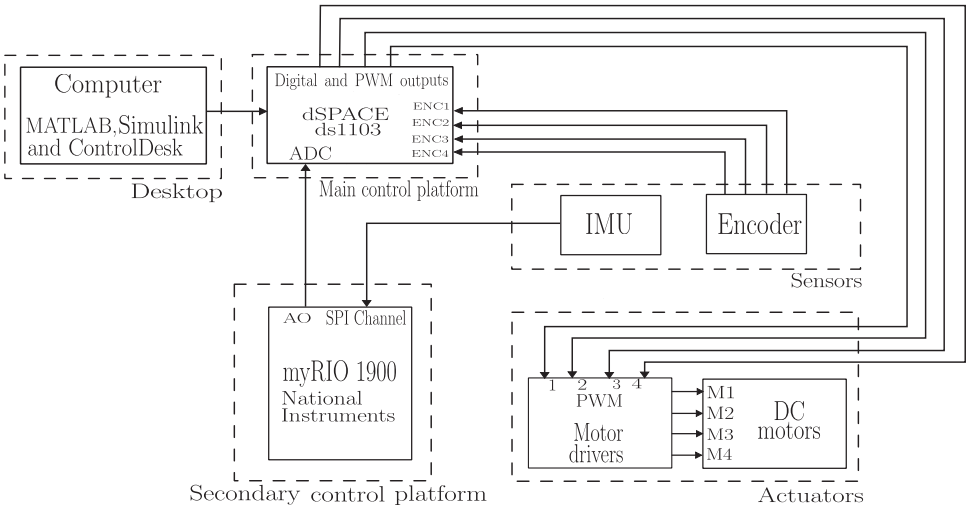


FIGURE 2 Data acquisition and control platforms scheme.

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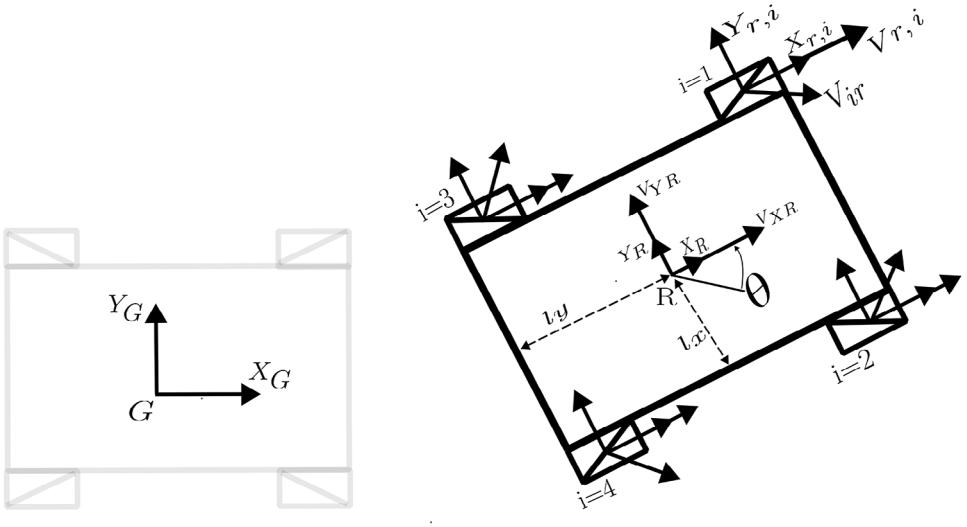


FIGURE 3 Kinematic entities of the mobile robot.

where T_ω is a transformation matrix that transforms the wheel velocities to the associated reference frame:

$$T_w = \begin{bmatrix} r & -\cos(\gamma_i) \\ 0 & \sin(\gamma_i) \end{bmatrix}.$$

Now, the velocities are projected from the reference frame of the ith wheel to the reference frame associated with the center of mass of the mobile robot through a pure z-axis rotation matrix representation:

$$\begin{bmatrix} V_{iX_R} \\ V_{iY_R} \end{bmatrix} = \begin{bmatrix} \cos(\phi_i) & -\sin(\phi_i) \\ \sin(\phi_i) & \cos(\phi_i) \end{bmatrix} \begin{bmatrix} V_{Xi} \\ V_{Yi} \end{bmatrix} = T_c T_w \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix}, \quad (4)$$

where T_c is a transformation matrix from the robot's reference frame X_r, r, Y_r to the reference frame associated with the robot's center of mass X_R, R, Y_R:

$$T_p = \begin{bmatrix} \cos(\phi_i) & -\sin(\phi_i) \\ \sin(\phi_i) & \cos(\phi_i) \end{bmatrix}. \quad (5)$$

In this case, the angle $\phi_i = 0 \ \forall \ i = 1, \ldots, 4.$ because there is no relative orientation between X_r and X_R. However, it is important to explicitly express this, as shown in Equation (5), because in a robot with a more complex geometric form, an angle ϕ_i may appear. Then, based on this explanation, we can rewrite Equation (4) as

$$\begin{bmatrix} V_{iX_R} \\ V_{iY_R} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_{Xi} \\ V_{Yi} \end{bmatrix} = T_c T_w \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix}. \quad (6)$$

Since the robot's movements occur in the plane, by recalling Equation (6), it is possible to express the global linear velocity of the mobile robot by taking into account the contribution of the angular or rotational velocity of the robot. Then, we have

$$\begin{bmatrix} V_{iX_R} \\ V_{iY_R} \end{bmatrix} = \begin{bmatrix} 1 & 0 & l_x \\ 0 & 1 & l_y \end{bmatrix} \begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_z \end{bmatrix} = T_g \begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_z \end{bmatrix}, \quad (7)$$

where T_g is a transformation matrix of the global velocities of the mobile robot in terms of the reference frame X_R, R, Y_R concerning the fixed reference frame X_G, G, Y_G:

$$T_g = \begin{bmatrix} 1 & 0 & l_x \\ 0 & 1 & l_y \end{bmatrix}.$$

Then, by relating Equation (3) to Equation (6), we obtain the following form:

$$T_c T_w \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix} = T_g \begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_z \end{bmatrix}. \quad (8)$$

3.3 | Direct and inverse kinematics of the mobile robot

With the form obtained in Equation (8), we have a first approach toward obtaining direct and inverse kinematics. On the one hand, direct kinematics allow us to determine the wheel velocities required to achieve a certain global velocity in the vehicle. On the other hand, with inverse kinematics, it is possible to determine the required global velocity of the robot. The direct kinematics are given by

$$\begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_z \end{bmatrix} = T_g^{-1} T_c T_w \begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix}, \quad (9)$$

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and the inverse kinematics by

$$\begin{bmatrix} \omega_i \\ V_{ir} \end{bmatrix} = (T_c T_w)^{-1} T_g \begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_{\tilde{z}} \end{bmatrix}. \tag{10}$$

Solving the matrix operations in Equations (9) and (10) yields the direct and inverse kinematic models in an explicit form for a rectangular mobile robot with four Mecanum wheels:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(l_x + l_y) \\ 1 & -1 & -(l_x + l_y) \\ 1 & -1 & -(l_x + l_y) \end{bmatrix} \begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_{\tilde{z}} \end{bmatrix}, \tag{11}$$

$$\begin{bmatrix} V_{XR} \\ V_{YR} \\ \theta_{\tilde{z}} \end{bmatrix} = \frac{r}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -\frac{1}{(l_x + l_y)} & \frac{1}{(l_x + l_y)} & -\frac{1}{(l_x + l_y)} & \frac{1}{(l_x + l_y)} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}. \tag{12}$$

4 | KINEMATIC CONTROL

With the information obtained from the kinematic model in the previous section, it is possible to identify that solving the position control problem in a mobile robot with Mecanum wheels also involves addressing control objectives in the wheel velocities. Based on this rationale, two control schemes are formulated: high-level control, which concerns position control, and low-level control, which concerns wheel-speed control.

4.1 | High-level control

4.1.1 | Definition of the high-level control problem

Consider the following reference position vector $q_r(t) = \begin{bmatrix} x_r(t) & y_r(t) \end{bmatrix}^T$, where $x_r(t)$ and $y_r(t)$ are the desired positions of the mobile robot in the plane. Then, the high-level control objective consists of formulating the control vector $V \in \mathbb{R}^2$ that reaches $\lim_{t\to\infty} \left\| q(t) - q_r(t) \right\| = 0$, where $q(t) = \begin{bmatrix} x(t) & y(t) \end{bmatrix}^T$ is the measured position vector obtained through the wheel velocity measurement, kinematic transformations, and numerical integrations, as we explain in previous sections. Figure 4 illustrates the motion control strategy based on kinematics and high- and low-level controllers. Different control techniques for high and low levels are presented below. Three well-known control techniques are explained below.

4.1.2 | Control 1. PID control

Despite the advances in control theory, PID control is the most widely explored controller. For this reason, a classical approach to the PID control law is

$$V(t) = K_p e_p(t) + K_i \int e_p(t)\, \mathrm{d}t + K_d \frac{e_p(t)}{\mathrm{d}t}, \tag{13}$$

where $V(t) \in \mathbb{R}^2$ is the input vector of control position signals, $K_p, K_i, K_d \in \mathbb{R}^{2\times 2}$ are diagonal gain matrices, and $e_p(t) \in \mathbb{R}^2$ is the error vector, where $e_p(t) = \begin{bmatrix} e_x(t) & e_y(t) \end{bmatrix}^T$; then, $e_x(t) = x(t) - x_r(t)$, and $e_y(t) = y(t) - y_r(t)$.

4.1.3 | Control 2. PID + sliding mode control

In a real-world context, the control problem is getting the system to track a specific time-varying desired state in unmodelled dynamics, as in this case study, where no dynamic model is considered. Then, classical approaches such as PID controllers cannot properly compensate for unmodelled dynamics. Another problem for classical approaches is hard nonlinearities, which may include Coulomb and viscous friction, saturation, dead zones, or a surface with a rough texture, which may cause the robot to experience slippage. Also, external disturbances may appear. Then, robust controllers must be developed. The so-called sliding mode control method is a simple approach to robust control consisting of a discontinuous function sign. Thus, we incorporate the sliding mode term into the standard PID controller in this study. Then

$$V(t) = K_p e_p(t) + K_i \int e_p(t)\, \mathrm{d}t + K_d \frac{e_p(t)}{\mathrm{d}t} - K_c \mathrm{sign}(e_p(t)), \tag{14}$$

where $K_c \in \mathbb{R}^{2\times 2}$ is a diagonal gain matrix for the robust component of the controller described by Equation (14) given by $\mathrm{sign}(e_p(t)) = \begin{bmatrix} \mathrm{sign}(e_x(t)) & \mathrm{sign}(e_y(t)) \end{bmatrix}^T$.

4.1.4 | Control 3. PID + quasi-sliding mode control

However, in practice, the discontinuous term of the sliding mode controller 14 exhibits a problem known as chattering, which produces audible noise and vibrations in mechanisms due to the high switching frequencies of the discontinuous control law, which is undesirable because it reduces the actuator's lifespan and can lead to irreversible damage to mechanical systems [37]. For these reasons, many explored strategies involve introducing a continuous approximation of the signum function while preserving the native robust characteristics of sliding mode control. The following approximation used for this study is [38]

$$\mathrm{sign}(e_p(t)) \approx \frac{e_p(t)}{e_p(t) + \varepsilon}, \tag{15}$$

where ε is a very small positive scalar. Thus, by substituting Equation (15) into Equation (14) we have

$$V(t) = K_p e_p(t) + K_i \int e_p(t)\, \mathrm{d}t + K_d \frac{e_p(t)}{\mathrm{d}t} - K_c \frac{e_p(t)}{e_p(t) + \varepsilon}. \tag{16}$$

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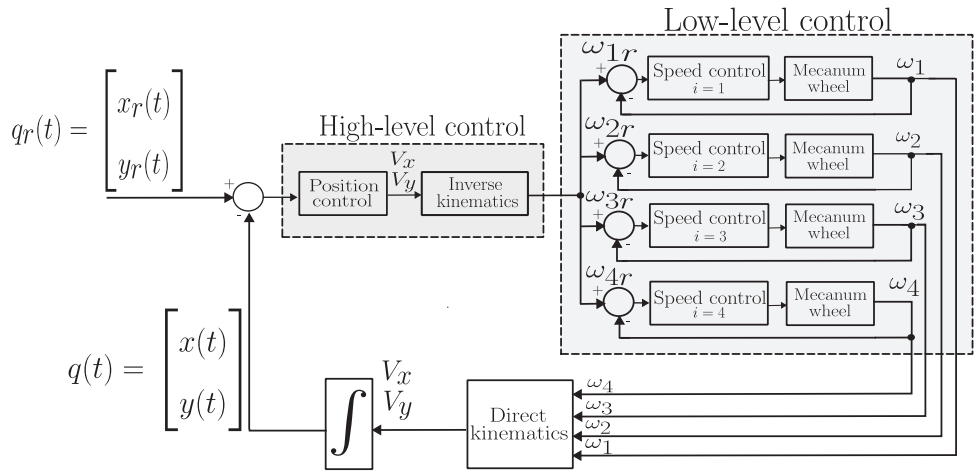


FIGURE 4 High-level and low-level control schemes employed for the mobile robot.

4.2 | Low-level control

4.2.1 | Definition of the low-level control problem

By introducing the control vector from the previous section, $V(t) = \begin{bmatrix} V_x(t) & V_y(t) \end{bmatrix}^T$, from the method referred to as high-level control into the inverse kinematic formulation described in Equation (10), it is possible to obtain the vector of desired angular velocities $\omega_r(t) \in \mathbb{R}^4$ for each Mecanum wheel. This can be interpreted as a signal transformation, transforming a high level into a low level. Then, the low-level control problem implies the design of vector control inputs $u(t) \in \mathbb{R}^4$ for each ith Mecanum wheel.

4.2.2 | Control 1. PI control

The low-level control vector is expressed as

$$u(t) = K_p e_v(t) + K_i \int e_v(t)\, \mathrm{d}t, \tag{17}$$

where $e_v(t)$ is the low-level error signal given by $e_v(t) = \omega - \omega_r(t)$. The subscript "$i$" is not expressed to avoid abuse of notation. However, the low-level control shown in Equation (17) is applied for each wheel.

5 | EXPERIMENT SETUP

In this section, we provide a detailed explanation of the experiment setup, which includes the methodology for measuring the interest variables and the proposed trajectories for trajectory tracking.

5.1 | Wheel speed measurement

The incremental optical encoder attached to each DC motor was employed for the angular velocity measurement of the wheels. After characterizing the encoder to convert its counts into angular position [rad] using the dSPACE ds1103 platform and Simulink programming, the angular velocity [rad/s] estimation starts by obtaining the difference between the angular position value at the previous sampling instance and the current sample. Then, the velocity is obtained by dividing this difference by the sampling time; the dSPACE ds1103 does this. This method for velocity estimation is often called the M method, according to the authors [39–41]. The *"Enc delta position"* block from the Simulink dSPACE library computes this position difference. Figure 5 illustrates the scheme for angular measurement, and the low-level control scheme and its main parts are explained as follows. The gain blocks convert encoder pulses into the wheel's angular position. Then, the result is divided by the sample time to obtain the angular velocity. The transfer function is a complementary filter for more reliable measurement and eliminates possible noise. The *"ur1"* input contains the low-level control signal, which is then converted into a PWM signal introduced to the motor drivers by the *"pwmr1"* output. The relational operators indicate the change in the direction of the motor concerning the value of the *"ur1"* signal. This scheme was repeated for each wheel.

5.2 | Position measurement

The position estimation was performed as explained below. With the angular velocity for each Mecanum wheel obtained, it is possible to ascertain the global velocities, measured by computing the kinematic transformation matrix from Equation (10). Subsequently, these global velocities were numerically integrated using the dSPACE ds1103 platform, with Euler 1 selected as the ordinary differential equation (ODE) solver and a fixed step of 0.001 s. Notably, the well-known issue of error accumulation did not arise primarily for the following reasons: 1. The experiments were conducted in an indoor laboratory environment, and the duration of each experiment was only 15 s; 2. The dSPACE ds1103 platform ensures real-time execution of control algorithms and provides greater reliability than

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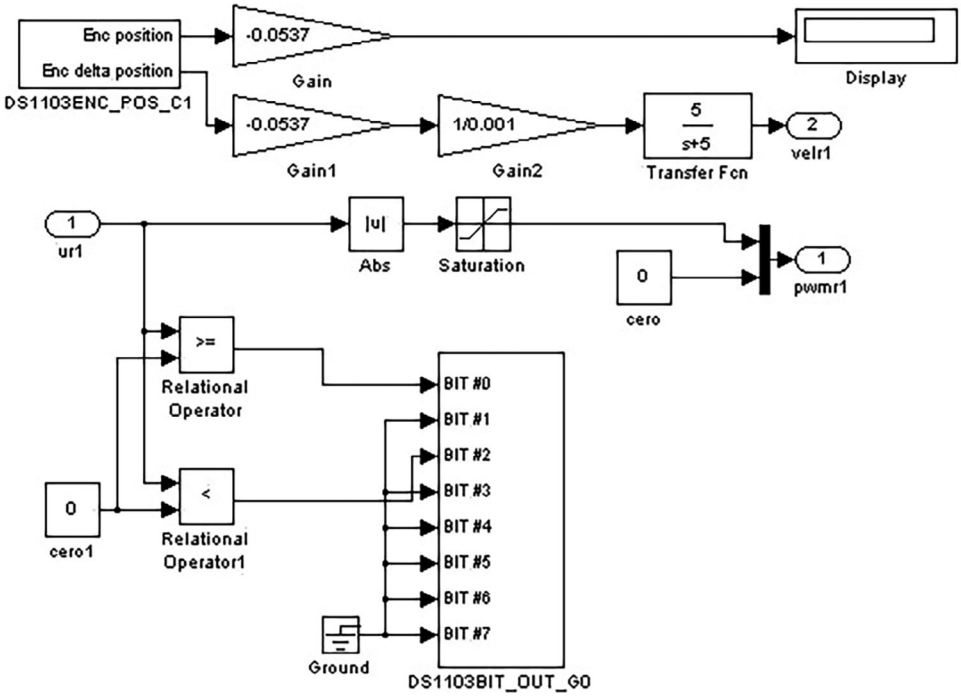


FIGURE 5 Low-level control signal implementation in Simulink software.

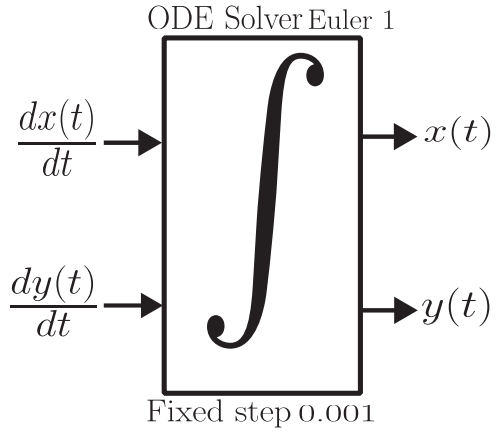


FIGURE 6 Numerical integration for position estimation during experimentation.

other control platform options, 3. Each experiment's integral action associated with position estimation was reset to eliminate cumulative errors. Figure 6 illustrates the method employed. However, at this point, it is important to say to the reader that for a large-scale application and long-term operation exposure, a more accurate and reliable method for position measurement needs to be employed, such as multi-sensor fusion [42], IMU and Kalman filter techniques [43], and the ultra-wideband (UWB) positioning system [44], among others [45].

5.3 | Reference signals for path tracking

The following two reference trajectories were selected. For the first test, a circular trajectory was proposed as the desired path, representing a simple and uniform motion, which can be considered a conventional trajectory and is given by the following reference signals: $x_r(t) = A\cos(\omega_f t)$, $y_r(t) = A\sin(\omega_f t)$. For the second test, a lemniscate-shaped reference was used to introduce a more complex and varied trajectory due to its curves, which can be considered complex. The reference signals are given by: $x_r(t) = A^2\cos(\omega_f t)$, $y_r(t) = \mathrm{A}\sin(2\omega_f t)$. The values for parameters were $A = 0.25$, and for the frequency $\omega_f = 2\pi f$, the value of $f = 0.125$ for both circular and lemniscate references.

6 | EXPERIMENTAL RESULTS AND COMPARISONS

This section presents the experimental results of implementing a kinematic control strategy on the mobile robot with Mecanum wheels. Greater emphasis will be placed on high-level control performance, as the goal is to control the robot's position. Two different surfaces were used to evaluate performance and demonstrate the strategy's applicability in relevant environments: the Laboratory floor, with a rough, dirty texture and thick tile joints, considered a conventional environment, and the second, a metal surface with skidding characteristics. Figure 7 illustrates both surfaces. A total of 120 experiments were conducted to obtain statistical indices that determine the repeatability of the experiments. The distribution of the experiments is explained as follows: for the PID control, 10 experiments were conducted on the laboratory floor and 10 on a metal surface for both a circular reference and a lemniscate trajectory reference. For the PID+quasi sliding mode control, 10 experiments were conducted on the laboratory

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FIGURE 7 The surfaces used for this experiment demonstration: laboratory floor, on the left, and metal surface, on the right.

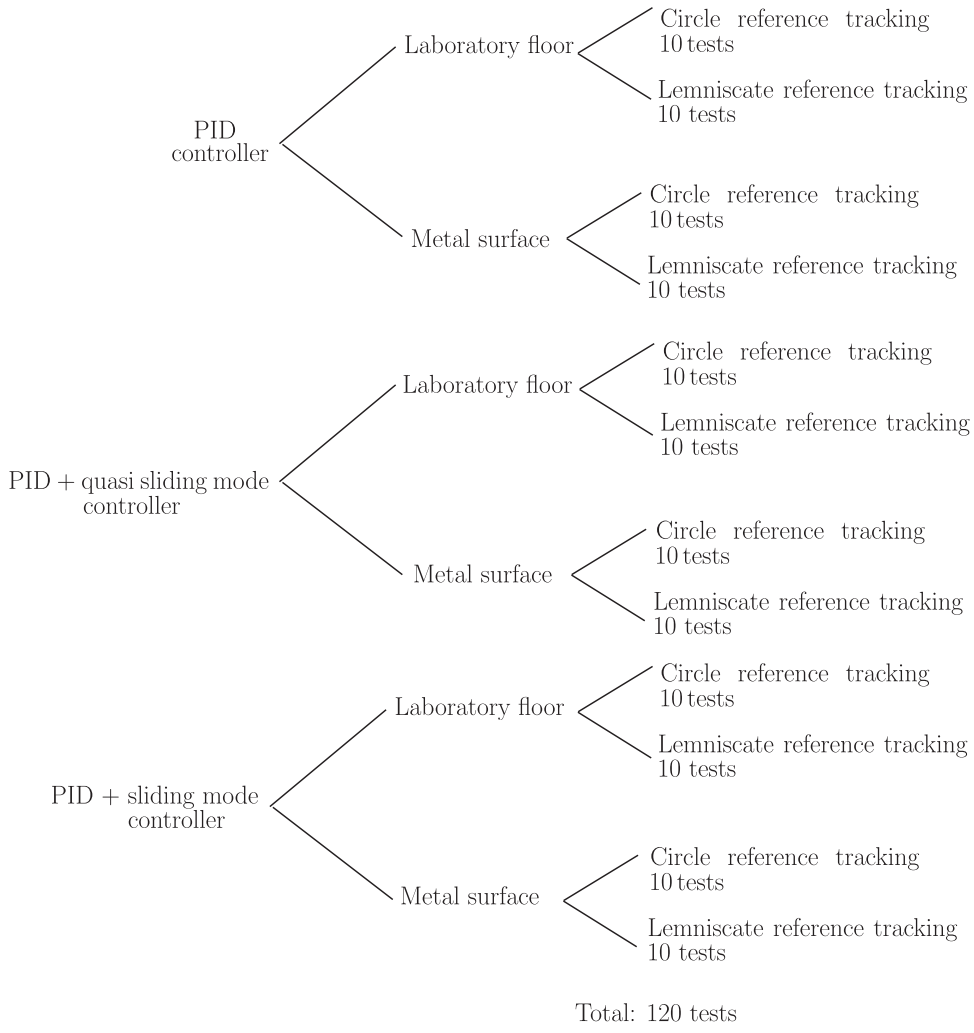


FIGURE 8 Tree diagram of the distribution of the 120 experiments developed.

floor, and 10 were performed on a metal surface for circular and lemniscate references, similar to the PID+sliding mode control. Figure 8 illustrates a tree diagram that compiles the 120 distribution experiments developed for this study. Table 5 presents the gains used for all experiments. The controller gains were tuned experimentally. For abbreviation purposes, please consider PID as proportional-integral-derivative, PID+SM as PID+sliding-mode control, and PID+QSMC as PID+quasi-sliding mode Control. This abbreviation will also appear later.

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TABLE 5 Gain utilized for experimentation.

Gain	PID	PID + SM	PID + QSM
K_p	5	5	5
K_i	0.5	0.5	0.5
K_d	2	2	2
K_r	—	0.1	0.1
ε	—	—	0.001

6.1 | Controller performance

Figures 9 and 10 illustrate the experimental results obtained for the circular tracking reference on the laboratory floor and the metal surface, respectively. Each graph contains ten overlapping experiments, resulting in 30 experiments per figure. This will also occur for the rest of the experimental Figures presented. The error signals $e_p(t)$ corresponding to the circular tracking references are illustrated in Figures 11 and 12 for the laboratory floor and metal surface, respectively. Similarly, the experimental results obtained for the lemniscate tracking reference are presented in Figures 13 and 14 for the laboratory floor and metal surface, respectively. The error signals $e_p(t)$ are illustrated in Figures 15 and 16 for the laboratory floor and metal surface, respectively.

Regarding the error signals $e_p(t)$ for both trajectories and both surfaces, illustrated in Figures 11, 12, 15, and 16, it can be observed that the signals quickly converge to zero, and their steady-state phase remains bounded. However, the PID + quasi-sliding mode controller exhibits better performance with the smallest enough bound than the other two proposed controllers, PID and PID + sliding mode. At this point, it is important to note that during experimentation, the robust control technique of PID + sliding mode generated significant vibrations in the vehicle, even though its gain was small. This is particularly evident in the lemniscate trajectory observable in Figures 13 and 14, which was the most complex

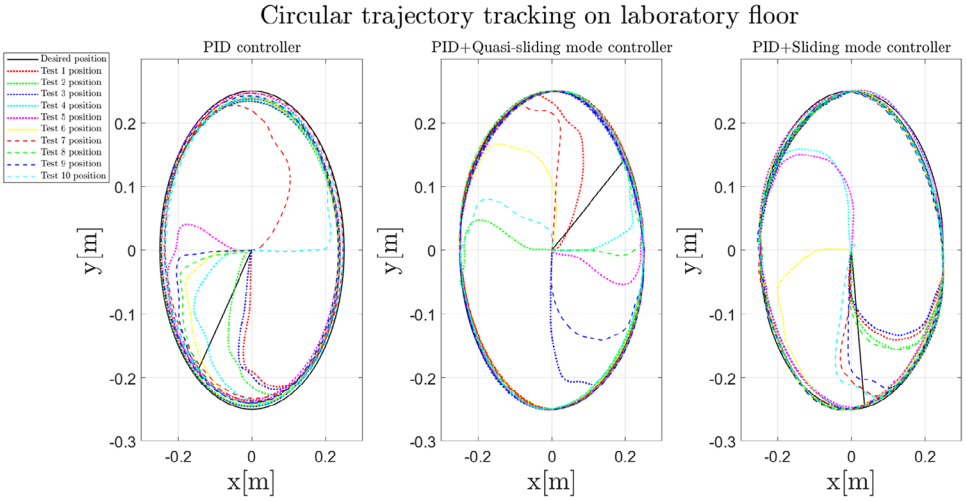


FIGURE 9 Performance of the mobile robot following a circular trajectory on the laboratory floor.

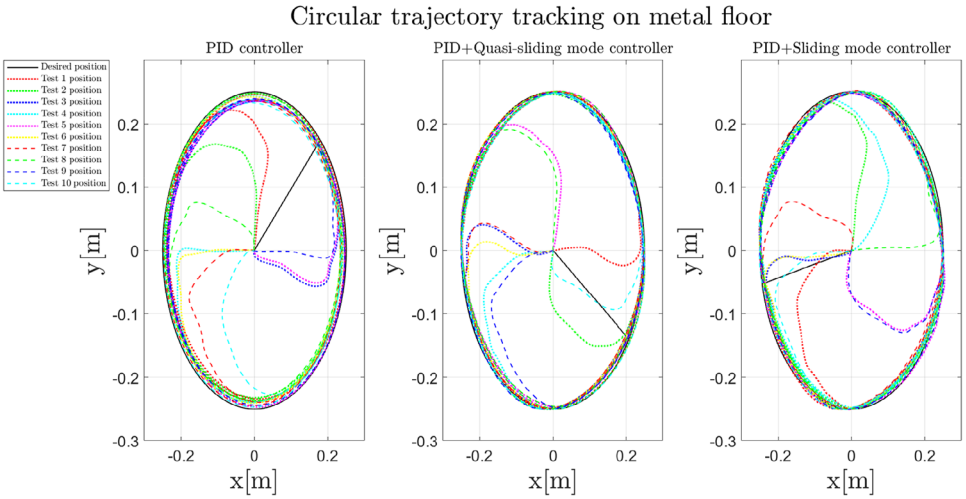


FIGURE 10 Performance of the mobile robot following a circular trajectory on the metal floor.

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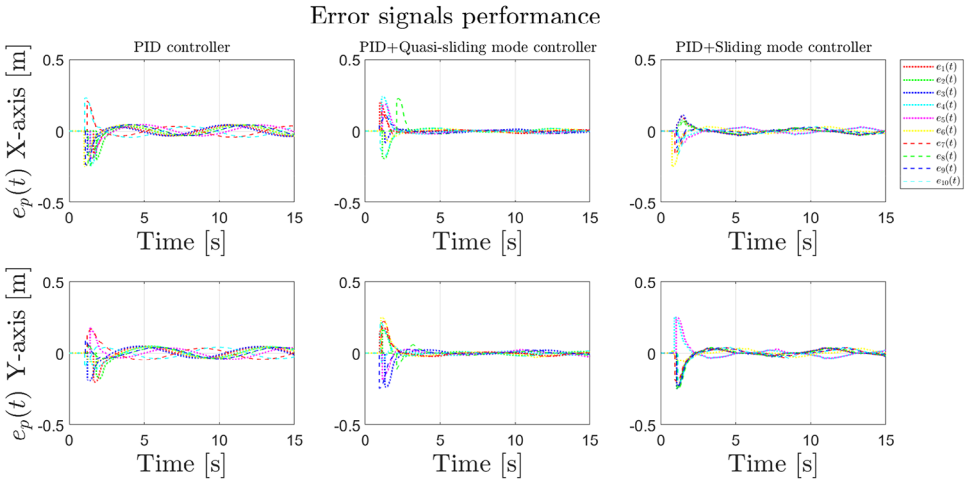


FIGURE 11 Error signals of the mobile robot's controllers during circular trajectory tracking on the laboratory floor.

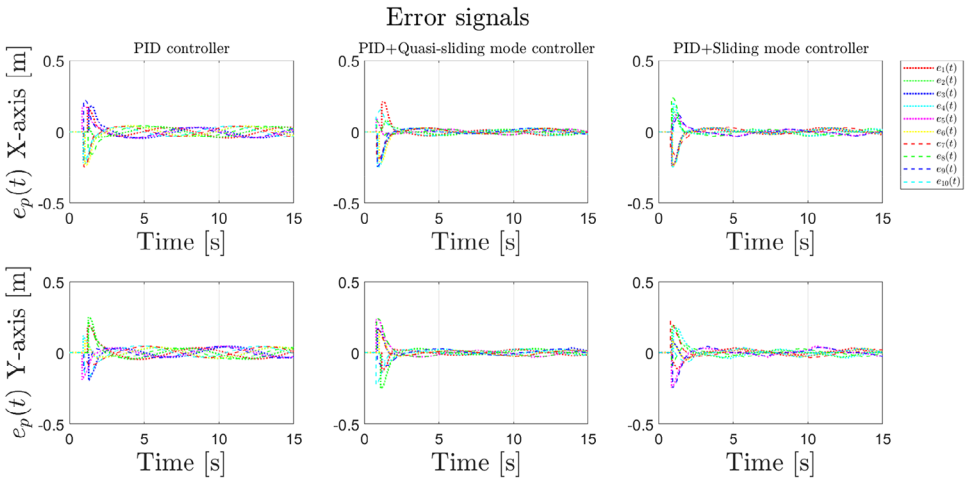


FIGURE 12 Error signals of the mobile robot's controllers during circular trajectory tracking on the metal floor.

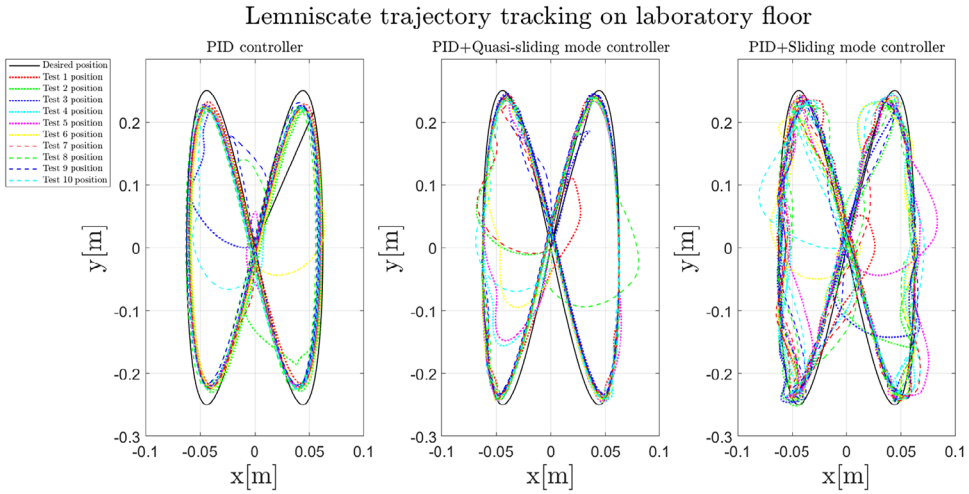


FIGURE 13 Performance of the mobile robot following a lemniscate trajectory on the laboratory floor.

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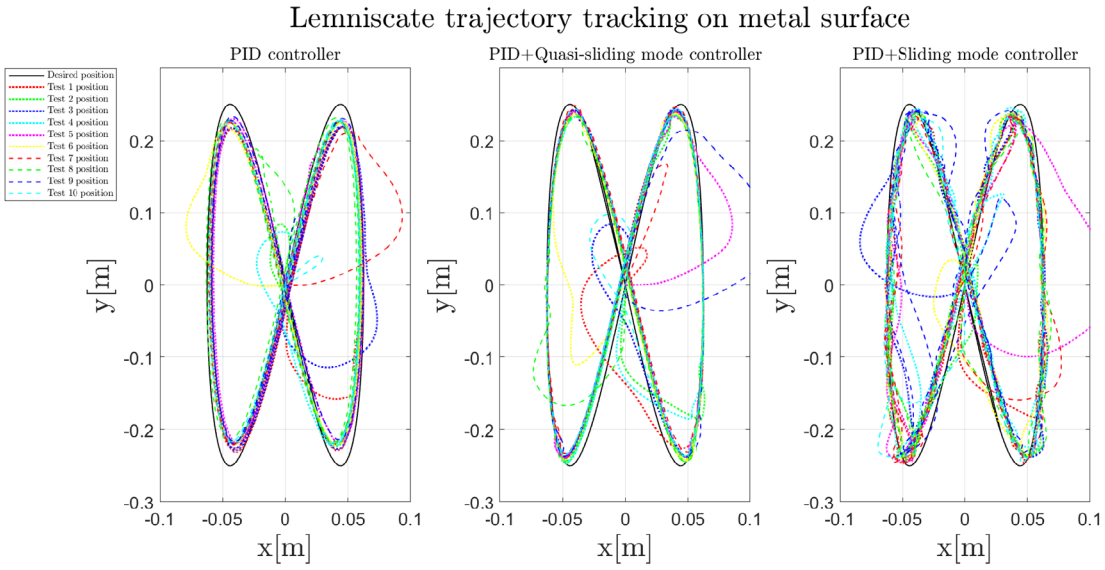


FIGURE 14 Performance of the mobile robot following a lemniscate trajectory on the metal floor.

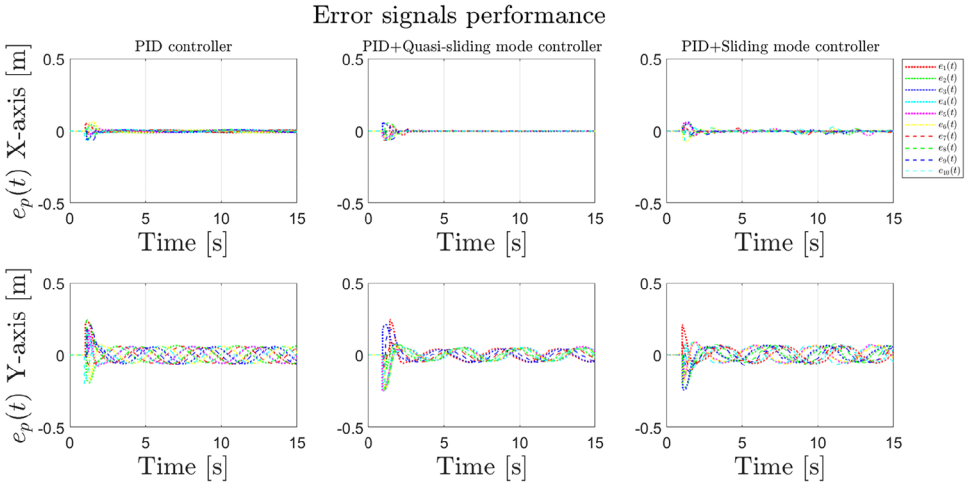


FIGURE 15 Error signals of the mobile robot's controllers during lemniscate trajectory tracking on the laboratory floor.

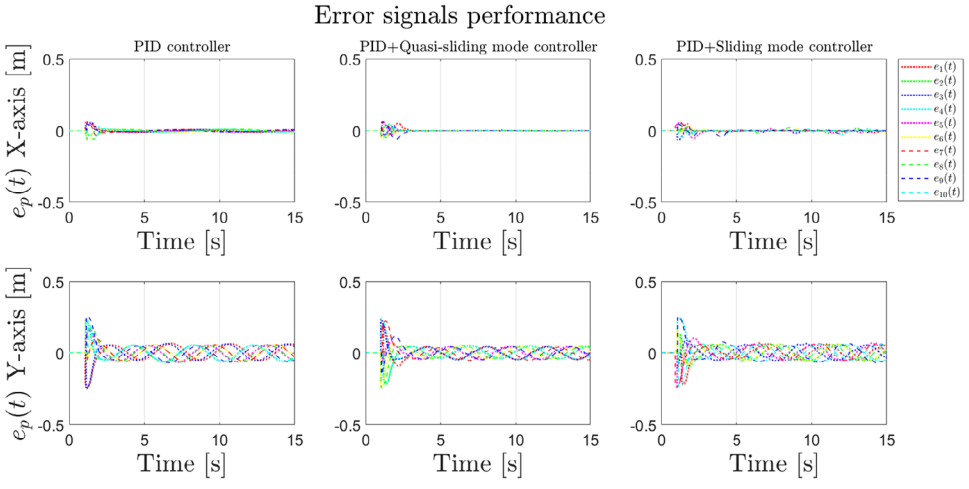


FIGURE 16 Error signals of the mobile robot's controllers during lemniscate trajectory tracking on the metal floor.

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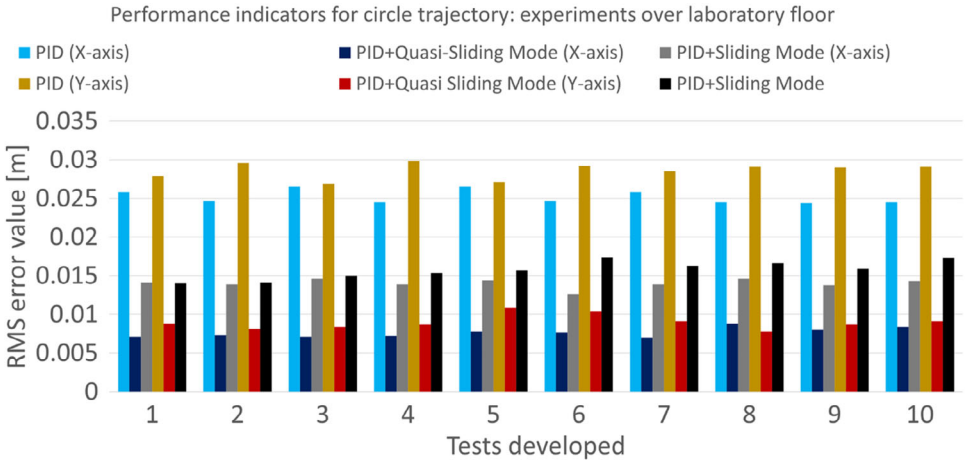


FIGURE 17 RMS values of tracking errors with circle reference trajectory on the laboratory floor.

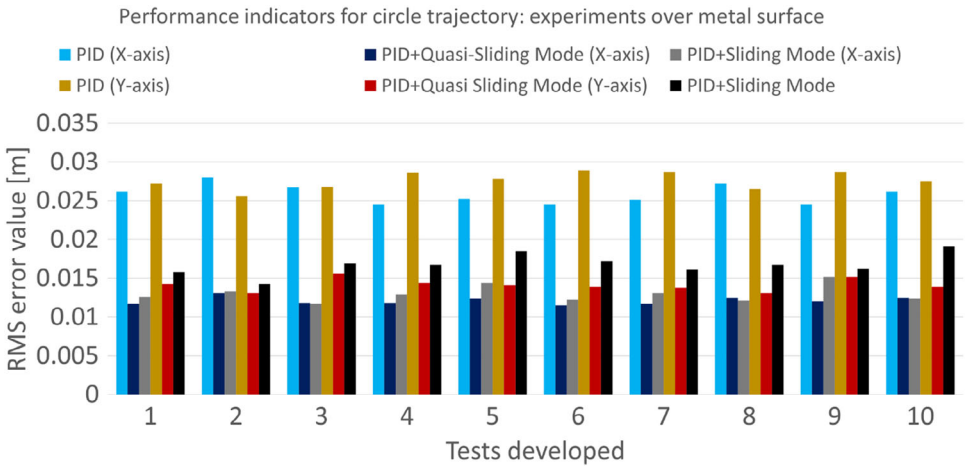


FIGURE 18 RMS values of tracking errors with circle reference trajectory on the metal surface.

trajectory. The vibrations, caused by the discontinuous term in the control signal, caused oscillations in the vehicle's trajectory tracking, resulting in poor performance. In this context, the proposal to use the widely explored PID controller combined with the continuous approximation of the sign function provided by quasi-sliding mode guarantees effective and desirable performance in practical scenarios for trajectory tracking of a Mecanum-wheeled robot.

On the other hand, to consider the results quantitatively and statistically, the following indices were used to evaluate steady-state performance

$$\|e\|_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|e\|^2},$$

where N is equal to the total samples. It is worth noting that given that the experiments conducted in this study were executed for 15 s, the statistical index $\|e\|_{\mathrm{RMS}}$ was computed after 5 s to evaluate the steady-state phase and omit transient responses, primarily because each experiment was executed independently, and the exact start time may vary. Figures 17–20 illustrates the RMS-error signal and the results are explained as follows. The RMS error signal obtained with the PID+quasi-sliding mode controller significantly differs from the other two implemented controllers. The percentage improvement was calculated by averaging the results obtained from tracking the RMS error values, and the results are listed below. For the circular trajectory on the laboratory floor, the PID + quasi-sliding mode controller showed an improvement of 69.99% for the X-axis and 68.6% for the *Y*-axis, both calculated relative to the standard PID control. Similarly, for the circular trajectory on the metal surface, the PID + quasi-sliding mode controller exhibited improvements of 53.1% and 48.9%, respectively. For the lemniscate trajectory on the laboratory floor, the improvement was 81.98% for the *X*-axis and 20.8% for the *Y*-axis. The same lemniscate trajectory on the metal surface demonstrated improvements of 87.4% and 24.5% for the *X*-axis and *Y*-axis, respectively, compared to the PID control. The percentage improvement results are more clearly illustrated in Table 6.

Regarding the low-level control, Figure 21 presents the performance of the vehicle's wheels in tracking the circular

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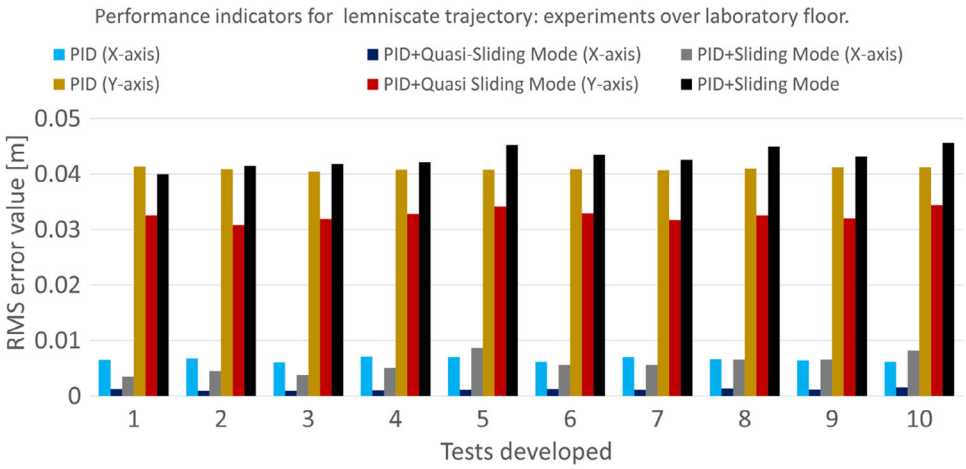


FIGURE 19 RMS values of tracking errors with lemniscate reference trajectory on the laboratory floor.

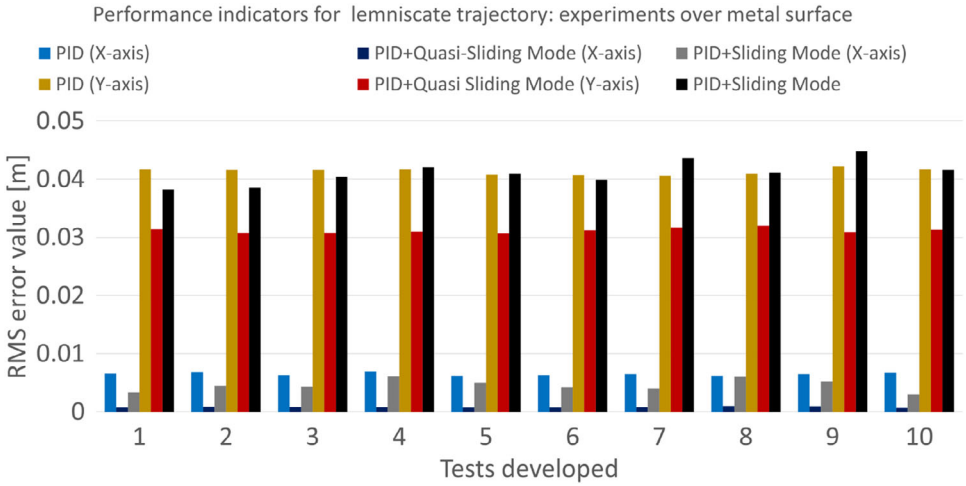


FIGURE 20 RMS values of tracking errors with lemniscate reference trajectory the metal surface.

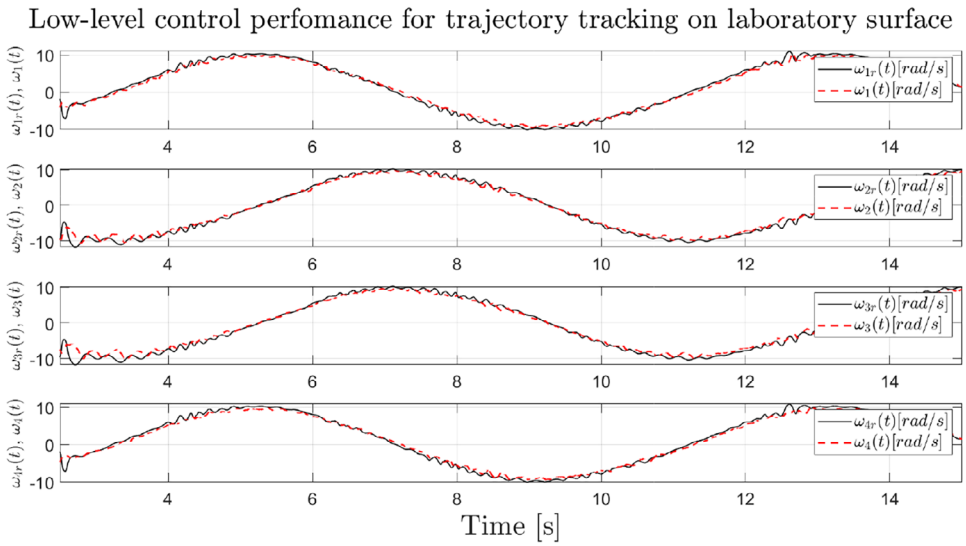


FIGURE 21 Low-level control performance: circular case, PID+quasi-sliding mode control on laboratory Floor.

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TABLE 6 Table with percentage improvement (%) of PID + quasi sliding mode against PID controller.

	Circle		Lemniscate	
Surface	Floor	Metal	Floor	Metal
X-axis	69.66%	53.10%	81.98%	87.40%
Y-axis	68.66%	48.90%	20.30%	24.50%

trajectory on the laboratory floor with the PID + quasi-sliding mode. The remaining experiments related to low-level control are not presented to avoid overuse and redundancy of figures. However, it is important to note the relationship between position control (identified in this study as high-level control) and low-level control, which is associated with regulating the speed of the robot's wheels to achieve the high-level objective. The control signals $u(t)$ are presented in Figures 22 and 23 for the circle trajectory on laboratory floor and metal surface, respectively, while Figures 24 and 25 for the lemniscate trajectory, on laboratory floor and metal floor, respectively. It is important to note the high frequencies and large amplitudes in the control signals associated with the PID+sliding mode control, which induce vibrations in the robot. On the other hand, the PID and PID+quasi-sliding mode controllers produce signals with less high-frequency components. The units of the signals are interpreted as PWM percentages.

For illustrative purposes, only the orientation angle $\theta(t)$ performance for the experiments following the circular and lemniscate trajectories on the laboratory floor is presented in Figure 26. It is crucial to highlight the orientation angle performance in mobile robots using Mecanum wheels, as the phenomenon known as slippage is one of the main factors leading to position control errors, according to [46]. In general, the issue of slippage leads to random and unexpected deviations in the orientation of the mobile robot during its movements, which is undesirable and contradicts the inherent advantages of an omnidirectional robot, whose key feature is the ability to move in any direction while maintaining its initial pose. In our experimentation, this problem also manifested in

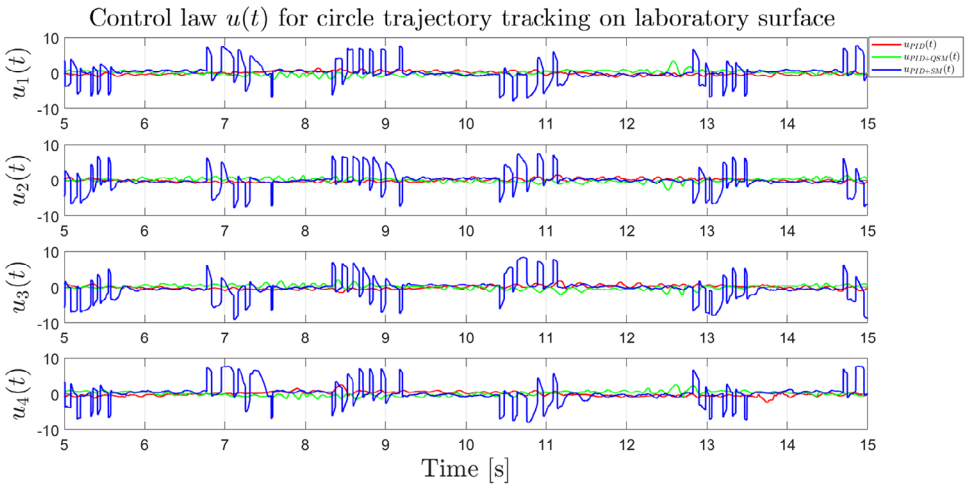


FIGURE 22 Control signals performance during the first test during circle trajectory tracking on laboratory floor.

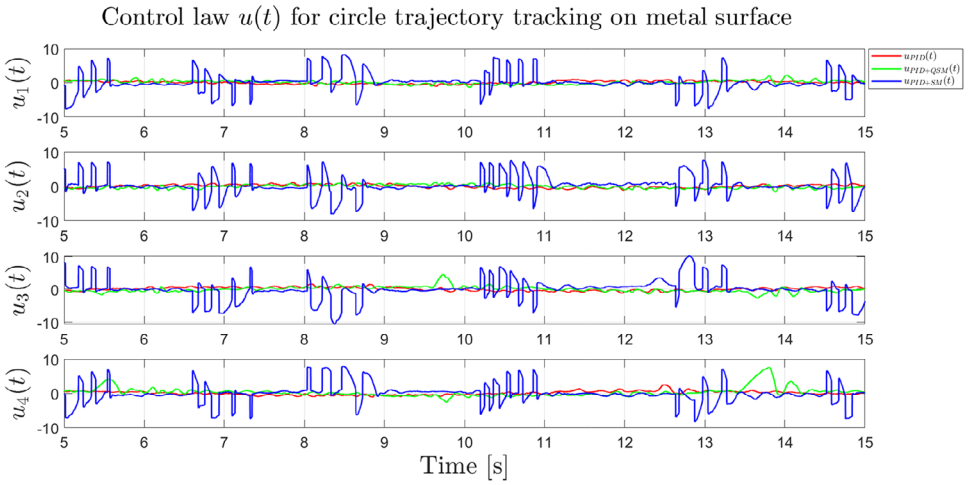


FIGURE 23 Control signals performance during the first test during circle trajectory tracking on metal surface.

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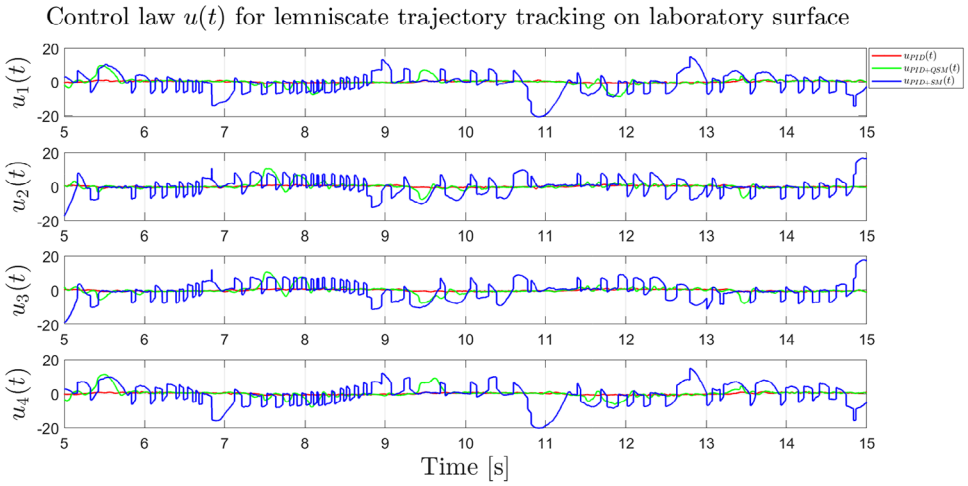


FIGURE 24 Control signals performance during the first test while lemniscate trajectory tracking on laboratory floor.

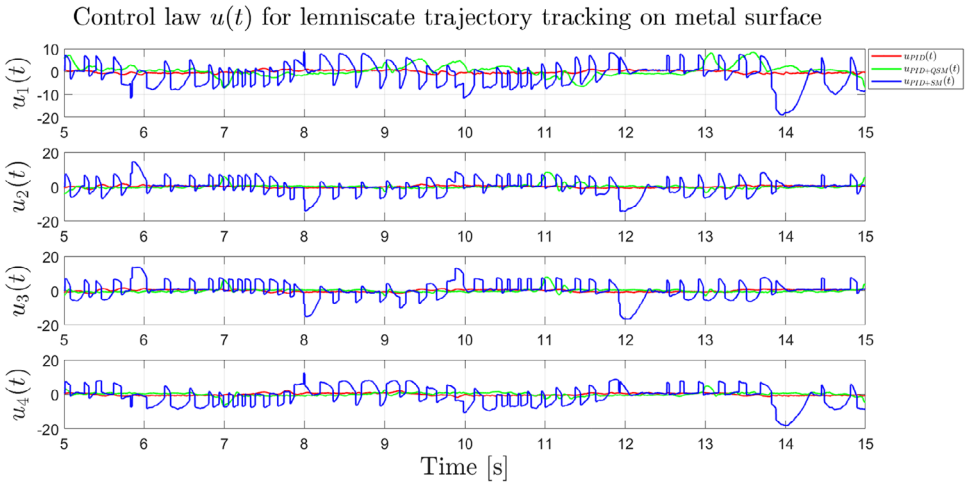


FIGURE 25 Control signals performance during the first test while lemniscate trajectory tracking on metal surface.

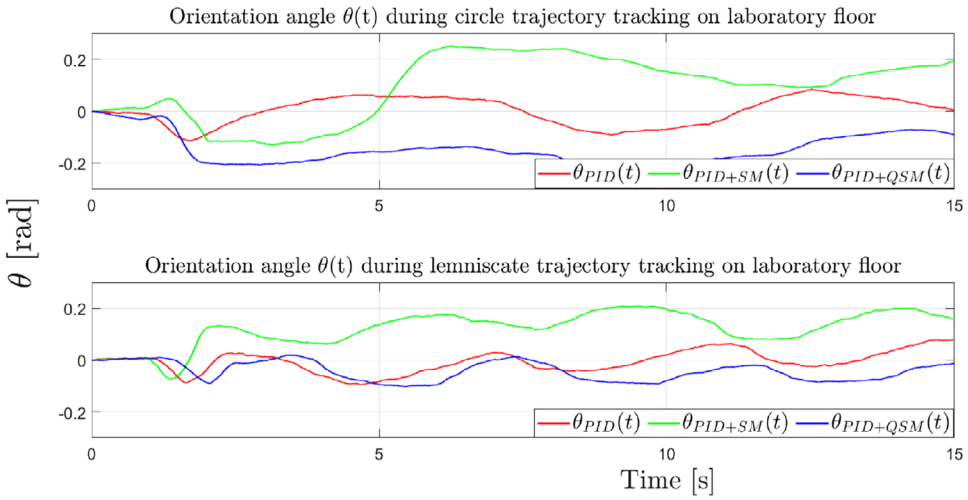


FIGURE 26 Orientation angle performance during the first experiment on the laboratory floor, circle, and lemniscate tracking trajectory.

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a similarly random manner. However, it is important to note in Figure 26 that the PID+sliding mode (PID+SM) control technique resulted in more significant deviations and oscillations in the vehicle's orientation compared to the other two methods, PID and PID+quasi-sliding mode (PID+QSM). This can be attributed to the vibratory effects caused by the high switching frequencies of the control law due to the discontinuous control part oscillating around +/- 2 radians, which is undesirable.

7 | CONCLUSIONS

This work addressed the kinematic control strategy in a Mecanum-wheeled mobile robot. It was demonstrated through 120 experiments that the control technique combining PID with quasi-sliding mode performs better than standard PID control or PID control with sliding mode, achieving significant differences. Kinematic control stands out as one of the initial approaches for navigation and path planning, in addition to being easy to implement and conceptually straightforward in design. One of the disadvantages of this strategy is that kinematic control does not account for the forces affecting the robot, such as friction, resistance to movement, external disturbances, or irregular surfaces, which could result in trajectories not being precisely followed in practice. Thus, in future work following this approach, we may use other robust controller techniques that do not depend on the dynamical model of the plant, like the active disturbance rejection control structure.

AUTHOR CONTRIBUTIONS

José Carlos Ortiz Hernández: Conceptualization; investigation; software; visualization; writing—original draft; writing—review and editing. **David I. Rosas Almeida**: Project administration; supervision; validation; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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